

## pyenv Python versions global, local and shell?

There are three types of Python versions under pyenv: global, shell and local. Please explain. Why we need to use this classification?



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Of course. This is an excellent question, as understanding these three levels is key to mastering `pyenv`. The classification exists to provide flexibility and precision in managing Python versions for different scenarios.

Here's a breakdown of the three types and why this hierarchy is necessary.

### 1. `pyenv global` - The Default Fallback

- **What it is:** This sets the default Python version for your entire user account on that MacBook. It's the version that will be used if no other specific version is set by the `shell` or `local` settings.
- **How it's set:** The version is stored in a file called `~/.pyenv/version`.
- **Command to set it:**

```
bash

pyenv global 3.11.6
```

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- **Why you need it:** It provides a sensible default. For example, you might set your `global` version to the latest stable release (e.g., `3.12.4`) so that any general scripting or tools you run outside of a project directory use a modern Python.

### 2. `pyenv local` - The Project-Specific Setting

- **What it is:** This sets a Python version for a specific project directory (and all its subdirectories). This is the most commonly used setting for development work.
- **How it's set:** It creates a file called `.python-version` in the current directory. The presence of this file tells `pyenv` to automatically switch to the specified version whenever you `cd` into that directory.
- **Command to set it:**

```
bash

cd ~/projects/my-legacy-app # Navigate to your project
pyenv local 3.8.18          # This creates the .python-version file
```

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### 3. `pyenv shell` - The Temporary Session-Specific Setting

- **What it is:** This sets a Python version for the current shell session **only**. It takes precedence over both `local` and `global` settings, but the setting vanishes as soon as you close the terminal window or run a new shell instance.
- **How it's set:** It sets the `PYENV_VERSION` environment variable for the current shell.
- **Command to set it:**

```
bash
```

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```
pyenv shell 3.10.13
```

- **Why you need it:** This is crucial for **reproducibility and isolation**. Different projects often have different requirements.
  - Project A might be an old Django app that only runs on Python 3.8.
  - Project B might be a new data science script that needs Python 3.12 with the latest features. Using `pyenv local`, you can `cd` into Project A and automatically have `python --version` report `3.8.18`, and then `cd` into Project B and have it report `3.12.4`. This prevents version conflicts and "works on my machine" problems.
- **Why you need it:** This is for temporary, one-off tasks. For example:
  - You want to quickly test a script against a different Python version without changing the project's `.python-version` file.
  - You're following a tutorial that requires a specific version for just a few commands.
  - It's a throwaway setting that you don't want to commit to your project's configuration.

### The Hierarchy of Precedence

This is the most important concept. `pyenv` checks these sources in the following order, and the **first** one it finds wins:

1. `PYENV_VERSION` environment variable (set by `pyenv shell`) - **Highest Priority**
2. `.python-version` file in the current directory (set by `pyenv local`)
3. `.python-version` file in the parent directories (searches upwards until it finds one)
4. `~/.pyenv/version` file (set by `pyenv global`) - **Lowest Priority**

You can visualize it like this:

```
shell -> local -> global
```

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### Why This Classification is Needed: The Analogy

Think of it like setting rules for where you can wear a hat:

- `global` : "In my house, the default is no hat." (The default rule)
- `local` : "But in my study room, everyone must wear a thinking cap." (A specific rule for a specific room/project)
- `shell` : "For the next hour, I will wear a party hat, regardless of which room I'm in." (A temporary, personal override)

This structure is needed because software development happens at different levels of specificity:

1. Your **machine** has a default ( `global` ).
2. Your **project** has a requirement ( `local` ).
3. You, the **developer**, might need to temporarily override everything for a quick test ( `shell` ).

Without this hierarchy, you would be constantly manually changing your `PATH` every time you switched projects, which is error-prone and incredibly inefficient. `pyenv` automates this context-switching, making your workflow seamless and reliable.



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